

VX/TPU — Change Log (V1.08-0)

Sector7 USA LLC · EDIT/TPU for Linux and Windows

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TPU Change Log

Product: Sector7 EDIT/TPU (`tpu.cpp`)

Current release: V1.08-0

Release date: 2026-07-24

Compared against: OpenVMS DECTPU **version=3 update=2** on host `t86` (OpenVMS x86_64 V9.2-3)

Release-numbering note: from this release TPU follows the same scheme as EDT — the middle field bumps once per release and the trailing edit field resets to 0. The earlier `V1.00-1` and `V1.00-2` builds are therefore renumbered here to `V1.01-0` and `V1.02-0`.

Bump `TPU_VERSION_*` / `TPU_RELEASE_DATE` / `TPU_SYSTEM_VERSION` / `TPU_SYSTEM_UPDATE` in `tpu.cpp` on every public release.

- **DECTPU-compatible:** `GET_INFO(SYSTEM, "version"|"update")` → integers
- **Sector7 extension:** `GET_INFO(0, "VERSION")` → `Vmajor.minor-edit` string

V1.08-0 — 2026-07-24

VMS `;n` file versioning

- **New: VMS-style file versioning, shared with EDT.** With versioning on, `tpu t.txt` opens the **highest existing version** (`t.txt;12`); `t.txt;1` opens version 1 (the bare `t.txt`); `t.txt;N` opens that version. Saving (Ctrl-S / `WRITE` / `SAVE` / `WRITE_FILE`) writes the next version: a brand-new file is written as the bare name (version 1), a bare name whose family exists writes `;highest+1` (`t.txt` → `t.txt;2`), and a `;N` name writes `;N+1` — never clobbering an existing version.
- **Turned on by** `TPU/VERSIONING` in the VX/RT config file (read directly from the file, not via `vxrt_config()`), a `SET VERSIONING ON` directive in `$HOME/TPU.INI` (honoured before the file is opened), the interactive `SET VERSIONING ON|OFF` command, or `S7SECURITY.INI` `versioning=on|off` as the fallback (default off). `SHOW VERSIONING` reports the state.
- A deployed VX/RT runtime (config found, or `libvxrtl.so` loads) suppresses the startup splash.
- Implemented via the shared `s7security/s7_vxconfig.hpp`, so EDT and VX-TPU cannot drift.

V1.07-0 — 2026-07-23

Startup splash

- **Fixed: the splash's blue background could leak into the editing screen.** Same latent defect as EDT V1.09-0: the splash sets a blue `bkgd()` and ncurses reuses the same screen for the editor, so the splash colour could persist. The background is reset when the splash is dismissed.

V1.06-0 — 2026-07-23

Shared-splash update, matching EDT (see the EDT changelog for the field feedback that drove it).

Startup splash

- **The initial splash no longer times out** — first run (and `splash=on`) waits for ENTER. Explicit `splash=<n>` / licensed durations remain timed.
- **The splash states it is shown once** and how to re-enable it.
- The shown-once notice and ENTER prompt anchor to the bottom of small terminals instead of being clipped; centering uses the real content height.
- Headless runs still start instantly (end of input dismisses).

V1.05-0 — 2026-07-21

Startup-splash polish, shared with EDT V1.07-0 via `s7security/`.

Startup splash

- **The splash now auto-dismisses.** Previously it blocked indefinitely waiting for RETURN, which meant the duration setting only decided *whether* it appeared, never for how long. It now honours a wall-clock timeout like EDT: 15 seconds by default, or whatever `splash=<n>` / the licensed duration says.
- **"Hit ENTER to continue"** — ENTER dismisses the splash early; other keys are ignored. The prompt wording changed from "Press RETURN" to match.
- **`splash=<n>` is clamped to 0..15 seconds** (was effectively unbounded up to one hour); out-of-range or negative values are treated as 15.
- **The persisted flag is now `splash=0`** (was `splash=off`); `off` is still accepted on read. `TPU_NO_SPLASH` still suppresses a single run without persisting anything.

V1.04-0 — 2026-07-20

Shared-splash update from the same field report that drove EDT V1.06-0.

Startup splash

- **`splash=<n>` sets the splash duration in seconds.** `S7SECURITY.INI` previously understood only on/off, so re-showing the splash meant deleting the file and catching it within the built-in 5 seconds. `splash=15` now holds it for 15 seconds; the count overrides both the licensed duration and the built-in `TPU_SPLASH_SECONDS` default. `splash=0` == `splash=off`, `splash=on` uses the default, and a keypress still dismisses early.
- Unrecognised `splash=` values are reported on stderr naming the value rather than being silently treated as "not off".
- Unchanged: the splash is shown **once**, then the file reverts to `splash=off`; `TPU_NO_SPLASH` still suppresses a single run without persisting anything.
- Shared with EDT via `s7security/` — same syntax, same precedence.

V1.03-1 — 2026-07-17

Test-isolation hotfix. The `v1.03-0` binaries were never shipped; this build is the first public `v1.03` release and carries the section/journal overwrite fix below (moved here from `v1.03-0`).

Fixes

- **`SAVE` / `JOURNAL_OPEN` force a clean overwrite** — both now `remove()` the target path before opening it for write. A stale section/journal file left in the working directory that is read-only or owned by another user (as happens on a shared home directory) could not be truncated in place, so the write failed silently and callers then read the stale content. This surfaced as two Linux-only (`rh8x`) regression failures — `EXPAND buffer` and `binary journal recovered (got="")` — while macOS passed because its scratch files were writable. Unlinking only needs directory write permission, so the file is replaced regardless of its old mode/owner.

- **Hermetic test harness** — `run_tests.sh`, `run_regression.sh` and `run_coverage.sh` now purge `tests/_tmp_*` scratch artifacts before running so a run never inherits a stale/non-writable section or journal file.

V1.03-0 — 2026-07-17

Licensing made optional + splash-once startup, from user field feedback. (Superseded by V1.03-1 before public release — no binaries shipped under this edit number.)

Licensing / startup

- **S7SECURITY.INI is now OPTIONAL** — TPU runs whether or not a security file is present. The editor no longer exits with `tpu: security file not found` when the file is missing or unreadable.
- **Splash shown once, then suppressed** — on first launch (no `splash=off` flag present) the branded splash shows once (dismissed with RETURN), after which TPU writes a plain-text `S7SECURITY.INI` containing `splash=off` at the resolved path (`$S7SECURITY`, else `$HOME/S7SECURITY.INI`). Subsequent launches read that flag and skip the splash. Delete the file (or set `splash=on`) to see it again. `TPU_NO_SPLASH` still suppresses for a single run without persisting the flag.
- **Legacy license blob still honoured** — an existing encrypted `S7SEC1` blob is read tolerantly to preserve the licensed name/email on the splash, but it is no longer required and is replaced by the plain `splash=off` flag once the splash has been shown. `s7security/mksecurity` and `s7licdb` still generate encrypted blobs (now effectively single-use).
- Unlicensed splash drops the "licensed to ..." line and shows the plain `(c) Sector7 USA LLC TPU` banner.

V1.02-0 — 2026-07-16

Splash + documentation fixes from user field feedback.

Fixes

- **Splash no longer flashes by** — `show_splash_screen()` now waits for the user to press RETURN instead of auto-dismissing after a short timeout, so it no longer vanishes before it can be read. It first discards input buffered before it draws (e.g. the Enter used to launch `tpu`) with `flushinp()` so it isn't dismissed instantly; a single RETURN then clears it immediately so it never lingers annoyingly.

Docs

- **PDF code blocks** — reference/changelog `pre` blocks now use a light background instead of dark navy, fixing the unreadable "dark bricks" in Adobe Reader and on hard copy.

V1.01-0 — 2026-07-15

Linux regression hardening (RHEL8 / `rh8x`) + build-script link fix + required S7SECURITY license.

Features

- **S7SECURITY license blob** — Shared with EDT (`s7security/`). SHA-256 authenticated (static linked); locate via `$S7SECURITY` or `~/S7SECURITY.INI`. Splash line `(c) Sector7 USA LLC TPU licensed to <name> <email>` when `seconds > 0`; **seconds == 0 skips the splash**. Create files with `s7security/mksecurity`.

Fixes

- **DELETE(buffer) undo Cleanup use-after-free** — statement-exit undo Cleanup no longer touches a freed `Buffer*` after `DELETE`; clears `snap_buf`, wipes buffer/marker/range bindings, and verifies the buffer is still in `editor.buffers` before pop/clear of undo stacks. Glitch manifested on glibc as corrupted size vs. `prev_size / SIGABRT` during headless suites (macOS was more forgiving).
- **Linux link:** `run_tests.sh / run_regression.sh / run_coverage.sh` add `-ldl` on Linux so `dlopen/dlclose (CALL_USER / SPAWN paths)` link cleanly on RHEL8 (plain `-lcurses` alone failed with `DSO missing from command line`).

Tests

- Core regression green on macOS and `rh8x`: **104 / 35 / 134** passed (`tpu_tests / tpu_complete / tpu_coverage`) + callable smoke.
- Stress: 10× each headless suite on `rh8x` with no heap aborts.
- `run_tests.sh` exports `S7SECURITY` to `s7security/testdata/S7SECURITY.INI`.

V1.00-0 — 2026-07-15

First numbered public release identity + OpenVMS fidelity pass.

Language / builtins

- `GET_INFO(SYSTEM, "version"|"update")` DECTPU form (integers)
- `GET_INFO(0, "VERSION"|"PLATFORM"|"RELEASE_DATE")` Sector7 extension
- `MODIFY_RANGE(range, start, end)` with `LINE_BEGIN` / `LINE_END` / `BUFFER_*` keywords
- `ERROR` / `ERROR_LINE` / `ERROR_TEXT` last-error tracking
- `MARK(FREE_CURSOR)` + `SET (FREE_CURSOR)` virtual columns past EOL
- `TRANSLATE` range form
- `WRITE_FILE(buffer, filespec)` and range form (DEC-compatible)
- `FILE_PARSE` / `FILE_SEARCH` / `EXPAND_NAME` real Unix paths
- `PROMPT([prompt] [, default] [, NOECHO|ECHO])`
- `EXPAND_LEARN(learn)` → program value
- `SCROLL` / `SHIFT` viewport forms; `SPAWN` Linux shell; `CURRENT_WINDOW` `WINDOW` value
- `CONSTANT` reassignment → `TPU$CONTRADEF`
- `ATTACH` keyboard forward (Ctrl-Z detach / Ctrl-D EOF)
- Kill ring + `PASTE PREVIOUS`; named pastes `STORE_PASTE` / `RECALL_PASTE`
- Appendix B `MESSAGE_TEXT` catalog (~325 symbols in `tpu_messages_appendix_b.inc`)
- `CLI /COMMAND=` / `/SECTION=` (+ `--` forms); `SHOW VERSION` / `SHOW PASTES`
- Pattern expression compile cache (≤256)
- App.C probes: `GET_INFO(..., "BOUND"|"BEFORE_BOL"|"BEYOND_EOL"|"MIDDLE_OF_TAB")`

Tests / OpenVMS

- Regression baseline at release: headless + callable (`./run_tests.sh`)
- Cross-check host `t86`: DECTPU `GET_INFO(SYSTEM, "version")=3, "update"=2`
- Probe scripts: `t86/run_probe.sh`, `t86/run_crosscheck.sh` (shell Task / `required_permissions: ["all"]`)

Intentionally deferred

- RMS / `;n` file versions
- DECwindows Motif widget tree (accepted no-ops)
- Byte-identical VMS `.TPU$SECTION` / keystroke journal layouts (host uses portable `#VXSECT` + TPJB instead)
- `CONVERT` / global select / Motif

Arrays (DEC GET_INFO walk)

- `CREATE_ARRAY(n [, start])` static portion; typed integer indexes

- `GET_INFO(arr,`
`"FIRST"|"NEXT"|"LAST"|"PREVIOUS"|"CURRENT"|"LOW_INDEX"|"HIGH_INDEX"|"SIZE")`
- `GET_INFO(ARRAY, ...)` global array list
- `arr{k} := UNSPECIFIED / TPU$K_UNSPECIFIED; DELETE(arr)` wipes refs
- Identifiers may contain `$` (facility constants)

Also in V1.00-0 tree (post-identity)

- Section `SAVE / RESTORE / EXPAND + /SECTION= (#VXSECT)`
- Binary `TPJB` journals (text still recoverable)
- Callable API: `tpu_api.h`, `libtpu`, `TPU_Initialize` / `TPU$INITIALIZE ...`

Pre-V1.00 — baseline

Unversioned Sector7 TPU: language + EVE layer, patterns, FAO, journals as text scripts, `--test` harness, ncurses/PDCurses.

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